

PEDRO H. R. DE ANDRADE

Computer Engineering
de Andrade, P. H. R.

Last updated: August 4, 2026

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ACADEMIC SUMMARY

I'm currently a Computer Engineering student with a passion for astronomy and data science. I have experience in scientific research, with a focus on computational astrophysics, and a history of publications in national conferences. I'm always seeking to learn and contribute to the scientific community, exploring new technologies and methodologies to advance our understanding of the universe. I have a hobby in animations and simulations, which allows me to create engaging visualizations to communicate complex concepts in an accessible way, with a focus on science outreach of physics-related content.

ACADEMIC BACKGROUND

Bachelor in Computer Engineering
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2023 – present Rio de Janeiro, Brazil

Technical Degree in Informatics
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
February. 2020 – December. 2022 Rio de Janeiro, Brazil

RESEARCH PROJECTS

Analysis of Dynamical Simulations of Merging Galaxy Clusters
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
September. 2022 – March. 2023 Rio de Janeiro, Brazil
Junior Scientific Initiation Scholarship - CNPq
Supervisor: Dr. Ana Cecília Soja

Simulating the Impact of Satellites on Astronomical Observations
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
2025 – present Rio de Janeiro, Brazil
Scholarship in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja

MobFog at IFFMaker
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2023 – February. 2024 Rio de Janeiro, Brazil
Scholarship in the Scientific Initiation project - IFF
Supervisor: Dr. Ana Cecília Soja

Anomaly Detection in Stars of the Milky Way: Exploring Data from Gaia and Other Surveys with Machine Learning
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
2024 – present Rio de Janeiro, Brazil
Scholarship in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja and Dr. Maria Luiza Linhares Dantas

EXTERNAL COURSES

Networks
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2020 Rio de Janeiro, Brazil

Operational Systems
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2021 Rio de Janeiro, Brazil

Server Administration
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2022 Rio de Janeiro, Brazil

RESEARCH INTERESTS

EVENTS

Science Fairs

-  **Analysis of dynamical simulations of merging galaxy clusters**
Poster – 21st Brazilian Science and Engineering Fair (FEBRACE)
2023 • São Paulo, Brazil
-  **Understanding dark matter from extragalactic shocks**
Poster – 40th International Science and Technology Fair (MOSTRATEC)
2023 • Novo Hamburgo, Brazil
-  **Simulating the impact of satellites on astronomical observations**
Poster – IX Brazilian Science Initiation Fair (FEBIC)
2024 • Pomerode, Brazil
-  **Anomaly detection in stars of the Milky Way**
Poster – XVI Scientific and Technological Exhibition of the Açaí Institute
2025 • Belém, Brazil

Conferences

-  **Analysis of dynamical simulations of merging galaxy clusters**
Oral Presentation – XV Scientific and Technological Initiation Congress and IX Fluminense Congress of Postgraduate Studies (CONFICT)
2024 • Campos dos Goytacazes, Brazil
-  **Analysis of dynamical simulations of merging galaxy clusters**
Oral Presentation – XVI Scientific and Technological Initiation Congress and X Fluminense Congress of Postgraduate Studies (CONFICT)
2025 • Campos dos Goytacazes, Brazil
-  **GALACTIC Catalog: stellar properties and chemistry**
Poster – XLVIII Annual Meeting of the Brazilian Astronomical Society (SAB)
2025 • Caxambu, Brazil
-  **GALACTIC Catalog: stellar properties and chemistry**
Oral Presentation – XVII Scientific and Technological Initiation Congress and XI Fluminense Congress of Postgraduate Studies (CONFICT)
2026 • Campos dos Goytacazes, Brazil
-  **Anomaly detection in stars of the Milky Way**
Poster – 78th Annual Meeting of the Brazilian Society for the Progress of Science (SBPC)
2026 • Niterói, Brazil
-  **Anomaly detection in stars of the Milky Way**
Poster – Winter School of Astrophysics
2026 • Rio de Janeiro, Brazil
-  **Anomaly detection in stars of the Milky Way**
Poster – XLIX Annual Meeting of the Brazilian Astronomical Society (SAB)
2026 • Natal, Brazil
-  **MobFog at IFFMaker**
Presentation – Congress of Teaching, Research and Extension (CONEPE)
2026 • Campos dos Goytacazes, Brazil

Academic Weeks

-  **Analysis of dynamical simulations of merging galaxy clusters**
Participant – VIII Knowledge Showcase and I Job Fair
2020 • Bom Jesus do Itabapoana, Brazil
-  **Understanding dark matter from extragalactic shocks**
Participant – IX Knowledge Showcase and II Job Fair
2021 • Bom Jesus do Itabapoana, Brazil
-  **Simulating the impact of satellites on astronomical observations**
Participant – I Week of Computer Engineering and Management

2021 • Bom Jesus do Itabapoana, Brazil

 Understanding dark matter from extragalactic shocks

Speaker – X Knowledge Showcase and III Job Fair

2022 • Bom Jesus do Itabapoana, Brazil

 Simulating the impact of satellites on astronomical observations

Participant – II Week of Computer Engineering and Management

2022 • Bom Jesus do Itabapoana, Brazil

 MobFog at IFFMaker

Poster – XI Knowledge Showcase and IV Job Fair

2023 • Bom Jesus do Itabapoana, Brazil

 MobFog at IFFMaker

Participant – III Week of Computer Engineering and Management

2023 • Bom Jesus do Itabapoana, Brazil

 MobFog at IFFMaker

Participant – XII Knowledge Showcase and V Job Fair

2024 • Bom Jesus do Itabapoana, Brazil

 MobFog at IFFMaker

Speaker and Organizer – IV Week of Computer Engineering and Management

2024 • Bom Jesus do Itabapoana, Brazil

 MobFog at IFFMaker

Participant – XIV Knowledge Showcase and VII Job Fair

2026 • Bom Jesus do Itabapoana, Brazil

Proceedings

[1] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *Boletim da Sociedade Astronômica Brasileira*. Vol. 37. 1. 2026, pp. 308–309.

[2] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « How varied are stellar populations in the Solar vicinity? Exploring GCNS and GALAH DR4 with t-SNE ». in: *Boletim da Sociedade Astronômica Brasileira*. Natal, RN, 2026.

[3] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Propriedades estelares e características químicas do Gaia Catalogue of Nearby Stars observado pelo GALAH DR4 ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2026.

[4] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Anais da 78ª Reunião Anual da Sociedade Brasileira para o Progresso da Ciência (SBPC)*. Niterói, RJ: UFF, 2026.

[5] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Resumos da Escola de Inverno de Astrofísica do Observatório Nacional*. Rio de Janeiro, RJ: Observatório Nacional, 2026.

[6] Maycon Jorge Deláqua da Silva, Pedro Henrique Rocha de Andrade, and Ana Cecília Soja. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2025.

[7] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *XVI Mostra de Ciência e Tecnologia do Instituto Ação (MCTIA)*. 2025.

[8] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e IX Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2024.

[9] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais da IX FEBIC – Feira Brasileira de Iniciação Científica*. 2024. ISBN: 978-65-991144-6-5.

[10] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. « Análise de simulações dinâmicas de aglomerados de galáxias em fusão ». In: *Anais da 21ª FEBRACE – Feira Brasileira de Ciências e Engenharia*. Vol. 21. 2023, p. 5. ISBN: 978-65-89190-18-9.

[11] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. « Entendendo a matéria escura a partir de choques extragaláticos ». In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 17.

[12] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Fabrício Barros Gonçalves. « Entendendo a matéria escura a partir de choques extragaláticos ». In: *Anais da MOSTRATEC 2023*. 2023, p. 102.

[13] Gabriel Coelho de Souza Poubel et al. « MobFog no IFFMaker ». In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 21.

Articles

[14] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *arXiv e-prints* (Dec. 2025). DOI: [10.48550/arXiv.2512.10037](#).

AWARDS

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IX FEBIC
3rd Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"
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XVI MCTIA
1st Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"

STATISTICS


3 Oral Presentations
National Events


7 Posters
National Events


9 Science Fairs/Congresses
National Events


10 Academic Weeks
Regional Events

PROGRAMMING

Python

git Git

Github

TeX

C++

Java

NodeJS

LANGUAGES

Portuguese


English


French


Spanish
